

TEMPORAL COHERENCE VIA VITERBI PATH SELECTION IN LATENT GRANULAR RESYNTHESIS

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ABSTRACT

Latent Granular Resynthesis encodes a source audio corpus and a target signal into the latent space of a neural audio codec, matches each target grain to its nearest codebook neighbour, and decodes the resulting hybrid sequence to produce audio that adopts the source’s timbre while preserving the target’s temporal structure. Because this matching is performed independently per grain, consecutive selections can be acoustically unrelated within the codebook, producing audible timbral discontinuities across grain boundaries. We propose Viterbi Grain Path Selection, a sequence constraint that penalises abrupt codebook transitions through a smoothness parameter λ . The method frames grain selection as a Hidden Markov Model inference problem and applies the Viterbi algorithm with a normalised cosine-distance transition matrix, where $\lambda = 0$ recovers the original greedy argmax exactly. As an ablation, we also evaluate soft latent blending, which replaces hard grain selection with a weighted convex combination of the top- k closest codebook grains. Both methods are evaluated on structural preservation via RMS envelope Pearson correlation across $\lambda \in \{0, 0.5, 1, 2\}$ and $k \in \{1, 3, 5, 10\}$. Viterbi path selection improves structural preservation over the greedy baseline, with λ controlling a smooth and interpretable trade-off between timbral fidelity and temporal continuity.

Index Terms— latent granular resynthesis, Viterbi decoding, neural audio codecs, timbre transfer

Code and audio examples availability:

<https://github.com/federico-de-lellis/viterbi-path-selection-latent-granular-synthesis>

1. INTRODUCTION

Granular synthesis has a history spanning over fifty years, rooted in Gabor’s theory of acoustical quanta and formalised by Roads [1] as a method of decomposing audio into short grains (typically 1–100 ms) that can be rearranged to create new sonic textures. Concatenative synthesis, or musical mosaicing [2, 3], extended this paradigm by automating grain selection through acoustic descriptor matching against a target signal. While powerful for timbre matching and cre-

ative sound design, these waveform-domain approaches suffer from audible discontinuities at grain boundaries due to the discrete nature of audio concatenation.

Neural audio codecs have recently transformed this landscape. Music2Latent [4], a consistency autoencoder that compresses audio into continuous 64-dimensional latent vectors at approximately 10 Hz frame rate, provides compact representations that implicitly interpolate during decoding. Tokui and Baker [5] exploited this property to introduce Latent Granular Resynthesis (LGR): a training-free method that encodes a source corpus into a codebook of latent grains, replaces each grain of a target signal with its cosine-nearest neighbour in the codebook, and decodes the hybrid sequence. Because the codec’s decoder upsamples and interpolates between grains, waveform-domain discontinuities are naturally eliminated.

However, the baseline LGR selects grains independently at each time step. This greedy, memoryless decision means that consecutive matched grains can be acoustically unrelated within the codebook, producing audible timbral jumps across grain boundaries even though the decoder smoothly interpolates each individual boundary. The problem is structural: no information flows between successive grain decisions.

This study introduces Viterbi LGR, a grain path through the codebook that penalises abrupt codebook transitions via a smoothness parameter λ . This formulation is analogous to how Hidden Markov Models [6] and attention-based sequence models [7] improved over greedy decoding in speech recognition and machine translation by considering global sequence structure. The Viterbi algorithm, applied here to codec latent space grain selection, finds the maximum-likelihood path through the codebook under both timbral similarity and transition smoothness constraints.

Secondarily we evaluate soft latent blending a top- k weighted convex combination of codebook grains as an intermediate ablation between hard selection and full optimal path. Both methods are assessed using RMS envelope Pearson correlation to measure structural preservation of the target signal.

2. RELATED WORK

Granular and concatenative synthesis. Roads [1] formalised granular synthesis as decomposing audio into short grains with quasi-Gaussian amplitude envelopes, establishing

the theoretical foundations for decades of subsequent work. Zils and Pachet [2] introduced musical mosaicing, framing the problem as constraint satisfaction over audio descriptors to automatically select and sequence grains from a database to match a target. Schwarz [3] surveyed the field comprehensively and identified the core tension between match quality and temporal continuity, the same tension motivating the present work. More recently, Tralie and Cantil [8] addressed continuity explicitly with a Bayesian particle filter for real-time concatenative synthesis, introducing a diagonal enhancement parameter that directly encourages temporal stability in grain selection.

Neural audio synthesis and timbre transfer. Bitton et al. [9] trained a Variational Autoencoder on individual sound corpora, enabling granular latent space sampling for novel sound generation. Pasini et al. [4] introduced Music2Latent, which provides a strong pre-trained foundation for training-free audio manipulation.

Latent Granular Resynthesis. Tokui and Baker [5] combined the granular paradigm with Music2Latent’s latent space, encoding a source corpus as a codebook of mean-pooled latent grains and matching each target grain via cosine similarity with a softmax temperature τ . The key limitation motivating our work is that grain selection is stationary and memoryless: each time step is optimised independently with no consideration of preceding or following selections.

Sequence decoding with global constraints. Rabiner [6] established the Viterbi algorithm as the canonical solution for recovering a state sequences under Markov transition constraints in Hidden Markov Models. Bahdanau et al. [7] and Vaswani et al. [10] showed that sequence modelling benefits from global context and non-greedy inference. Our work applies both hard (Viterbi path) and soft (top- k blending) variants of these ideas to neural audio codec grain selection.

3. METHODOLOGY

3.1. Baseline: Latent Granular Resynthesis

The baseline method operates in three stages. First, the source audio corpus is encoded through Music2Latent, producing a latent tensor of shape $(T, 64)$ at approximately 10 Hz frame rate. This tensor is segmented into grains of `grain_size` consecutive frames with a step of `stride` between successive grain start positions. Each grain is represented both by its mean-pooled vector (for similarity matching) and its original multi-frame sequence (for reconstruction). Pitch-shift augmentation at ± 1 and ± 2 semitones expands the codebook’s timbral coverage.

Second, the target audio is encoded and segmented identically. Each target grain is matched to its closest codebook grain via cosine similarity. A softmax temperature τ controls selection. The selection probability is:

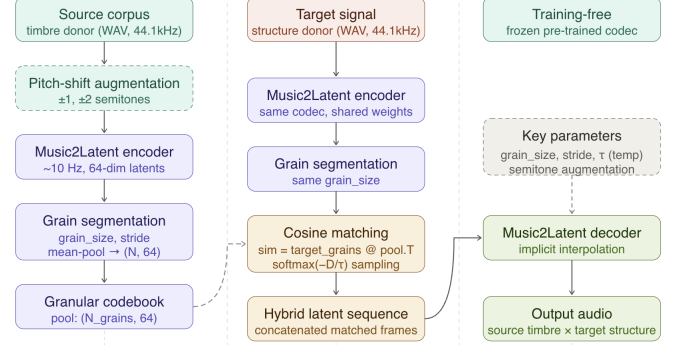


Fig. 1. Latent Granular Resynthesis (LGR) pipeline implemented in this study.

$$P(\text{select } B_i) = \frac{\exp(-D_{\cos}(A, B_i)/\tau)}{\sum_j \exp(-D_{\cos}(A, B_j)/\tau)} \quad (1)$$

where $D_{\cos}(A, B_i) = 1 - \cos(A, B_i)$.

Third, the matched grain frame sequences are concatenated into a hybrid latent tensor and decoded through Music2Latent’s decoder.

3.2. Viterbi Grain Path Selection

We formulate grain selection as a Hidden Markov Model inference problem. Codebook grains are hidden states, target grains are observations, and Viterbi decoding recovers the globally optimal path under emission (timbral similarity) and transition (codebook smoothness) constraints.

Emission matrix. $E \in \mathbb{R}^{N_t \times N_p}$ stores cosine similarity between each target grain and each pool grain.

Transition cost matrix. $\text{Tr}[i, j] = 1 - \cos(g_i^{\text{pool}}, g_j^{\text{pool}})$ measures pairwise dissimilarity between pool grains.

Transition normalisation. The implementation rescales transition costs to match the emission scale:

$$\text{Tr}^* = \text{Tr} \cdot \frac{\bar{E}}{\bar{\text{Tr}}} \quad (2)$$

Forward pass.

$$V[t, j] = \max_i (V[t-1, i] - \lambda \cdot \text{Tr}^*[i, j]) + E[t, j] \quad (3)$$

with backpointer

$$\text{ptr}[t, j] = \arg \max_i (V[t-1, i] - \lambda \cdot \text{Tr}^*[i, j]) \quad (4)$$

and initialisation $V[0, j] = E[0, j]$.

Backtracking. The optimal path is recovered as

$$\text{path}[N_t - 1] = \arg \max_j V[N_t - 1, j] \quad (5)$$

then recursively via stored backpointers.

When $\lambda = 0$, Viterbi reduces exactly to the greedy argmax baseline. Increasing λ encourages smoother trajectories at the cost of per-grain timbral fidelity.

3.3. Soft Latent Blending (Ablation)

As an ablation, we evaluate soft latent blending as an alternative to hard grain selection. Instead of selecting one grain per target grain, we take the top- k nearest grains by cosine similarity and computes a weighted convex combination of their frame sequences. The softmax weights over the top- k similarities are given by:

$$w_i = \frac{\exp(\cos(g_t, g_i^{(k)})/\tau)}{\sum_{j=1}^k \exp(\cos(g_t, g_j^{(k)})/\tau)} \quad (6)$$

The blended grain is then:

$$\hat{g}_t = \sum_{i=1}^k w_i \cdot G_i^{\text{frames}} \quad (7)$$

That denotes the multi-frame tensor of the i -th closest pool grain. This blended tensor is passed to the decoder in place of a single grain’s frames. This probes whether convex combinations in Music2Latent space decode to coherent audio.

3.4. Evaluation Metrics

We evaluate structural preservation as the Pearson correlation between RMS energy envelopes of target and hybrid signals (frame_length=2048, hop_length=512, 44.1 kHz).

4. EXPERIMENTAL SETUP

All experiments use Music2Latent [4] at 44.1 kHz, latent frame rate approximately 10 Hz, and 64-dimensional latent vectors. The source audio and the target can be chosen, both mono 44.1 kHz WAV files. The codebook uses `grain_size` that carry different temporal meaning in different configuration are suggested and `stride` where smaller values allows greater coverage through overlapping segments instead larger stride forces more diversity between grains. Pitch-shift augmentation at -2 , -1 , $+1$, and $+2$ semitones yields five source variants.

The Viterbi sweep uses $\lambda \in \{0.0, 0.5, 1.0, 2.0\}$, with $\lambda = 0$ as greedy baseline. The soft blending ablation uses $k \in \{1, 3, 5, 10\}$ and $\tau = 1.0$. Method are evaluated using RMS envelope Pearson correlation.

5. RESULTS AND DISCUSSION

Grain usage. The Viterbi path concentrates selections on a smaller subset of grains, reflecting the transition constraint’s preference for local continuity.

Transition normalisation. Without normalisation, transition costs dominate emissions and collapse the path to near-constant states for $\lambda > 0$. With normalisation, λ becomes more interpretable.

Grain size considerations. The grain size with high values is coarser than classical granular synthesis [1], but aligns with Music2Latent’s 10 Hz latent framing and keeps the transition matrix tractable especially with longer tones.

6. CONCLUSION

We presented an extension of the approach proposed by Tokui and Baker [5] on the grain selection still as a training-free global sequence constraint for latent granular resynthesis as new technique that uses neural audio codecs for creative granular resynthesis, allowing for high-quality, and quite flexible timbre transfer without the need for additional training. The smoothness parameter λ is interpretable: $\lambda = 0$ reproduces greedy argmax exactly, while larger values is an attempt to produce smoother trajectories and improved structural preservation up to a plateau.

Current limitations include the reliance on a single active quantitative metric and the lack of a formal and deep listening study.

Future work includes approximate or chunked Viterbi decoding for large codebooks and streaming, activation of timbre similarity metrics, perceptual evaluation across settings, and validation across additional codec backbones.

By working within the latent space of pre-trained codecs, the method enables smooth blending of timbres while maintaining the structural features of the target audio.

The main advantage of the implementation is its simplicity and ease of use, requiring only a source dataset and a target signal to produce rich hybrid sounds. The flexibility to incorporate varied source materials and control the level of granularity offers both artists and researchers a powerful framework for creative experimentation.

7. REFERENCES

- [1] Curtis Roads, “Introduction to granular synthesis,” *Computer Music Journal*, vol. 12, no. 2, pp. 11–13, 1988.
- [2] Antoine Zils and François Pachet, “Musical mosaicing,” in *Proceedings of the International Conference on Digital Audio Effects (DAFx)*, Limerick, Ireland, 2001.
- [3] Diemo Schwarz, “Concatenative sound synthesis: The early years,” *Journal of New Music Research*, vol. 35, no. 1, 2006.
- [4] Michele Pasini, Stefan Lattner, and György Fazekas, “Music2latent: Consistency autoencoders for latent audio compression,” in *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2024.

- [5] Nao Tokui and Tom Baker, “Latent granular resynthesis using neural audio codecs,” in *ISMIR Late-Breaking Demo Session Extended Abstracts*, 2025.
- [6] Lawrence R. Rabiner, “A tutorial on hidden markov models and selected applications in speech recognition,” *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.
- [7] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio, “Neural machine translation by jointly learning to align and translate,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2015.
- [8] Christopher Tralie and Bertil Cantil, “The concatenator: A bayesian approach to real-time concatenative musaicing,” in *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2024.
- [9] Ari Bitton, Philippe Esling, and Tatsuya Harada, “Neural granular sound synthesis,” in *Proceedings of the International Computer Music Conference (ICMC)*, 2021.
- [10] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.